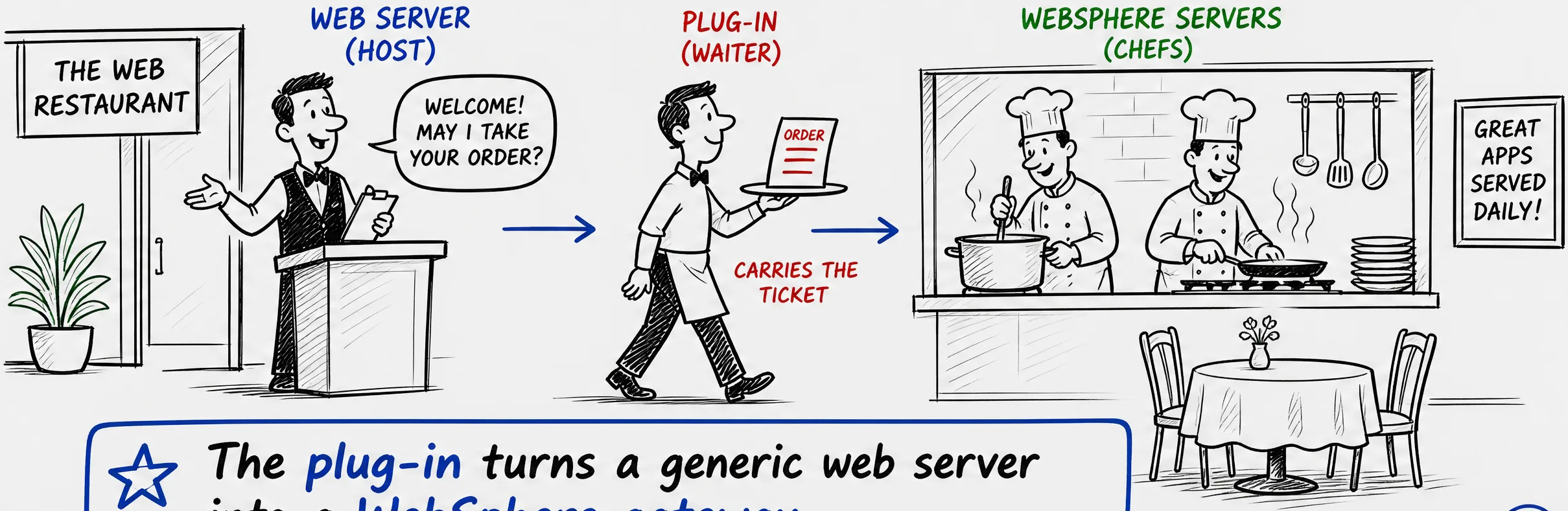
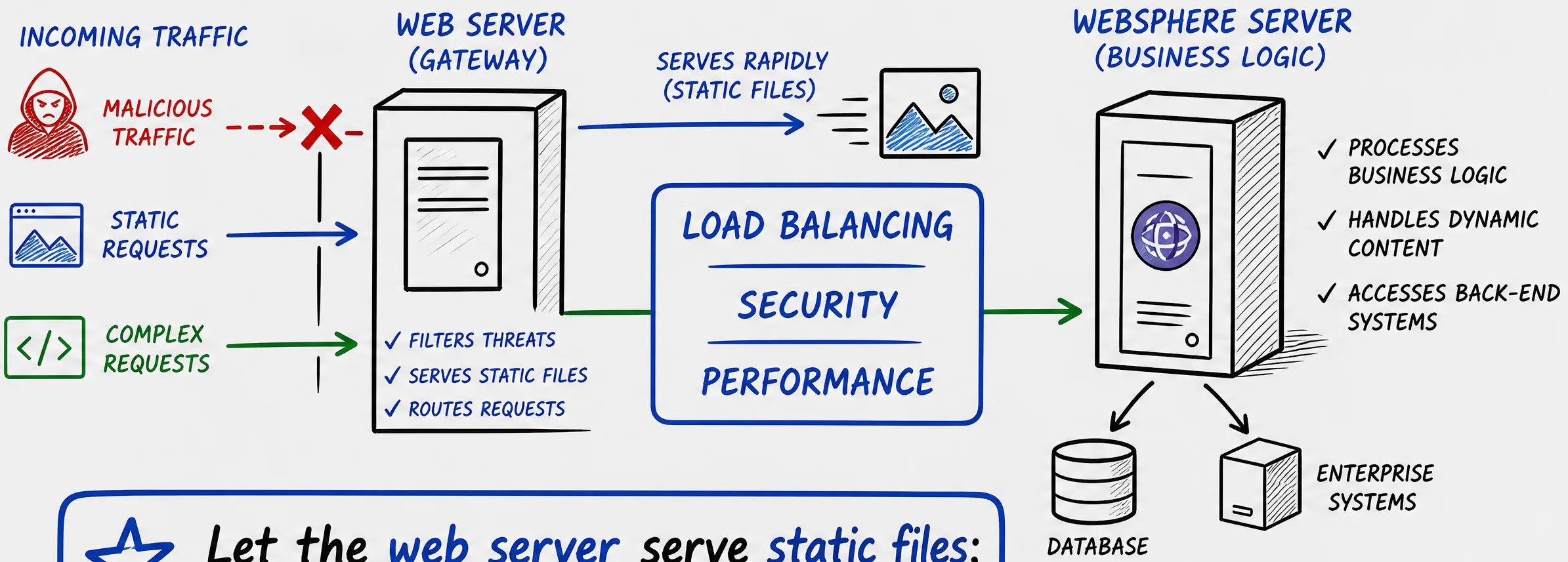
IBM HTTP Server and Plug-in



☆ The plug-in turns a generic web server into a WebSphere gateway.

Why Use a Web Server?

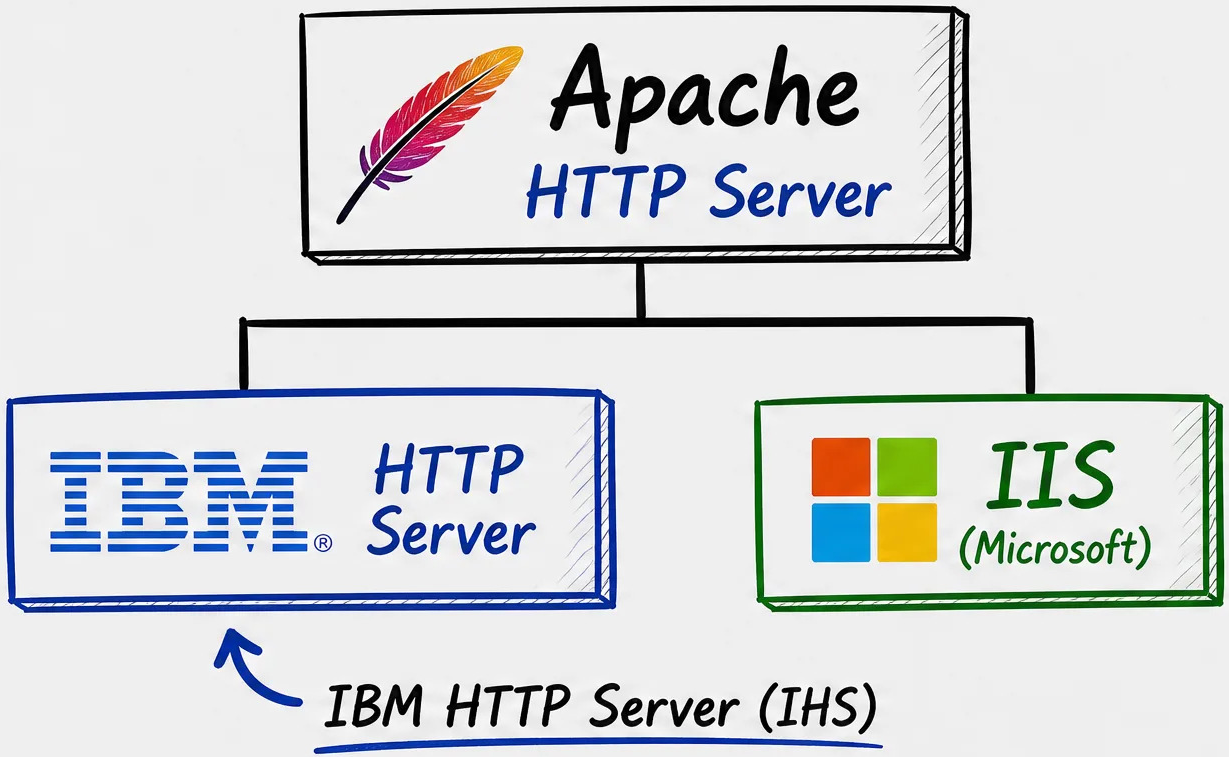


Let the web server serve static files; let WebSphere serve business logic.

ProTech
protechtraining.com

Supported Web Servers

Dr. Lee



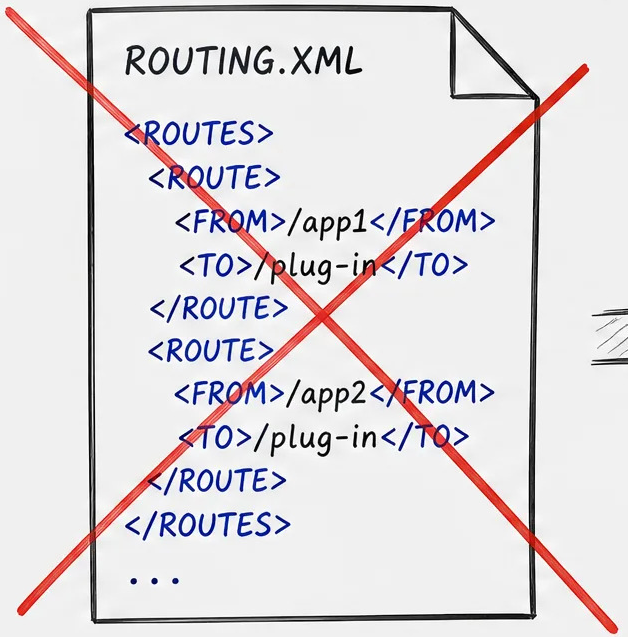
Built on
Apache

IHS speaks WebSphere's language natively.

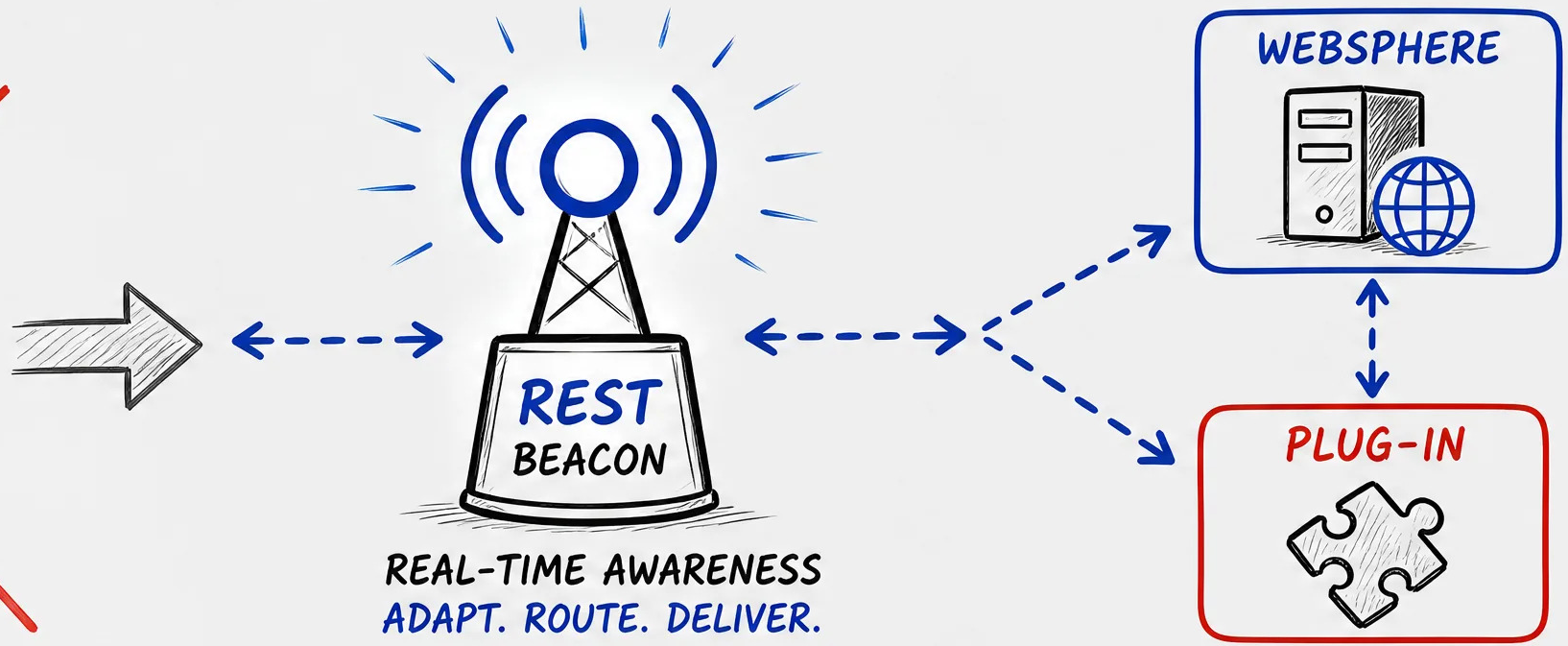


Intelligent Management (New in v9)

OLD WAY:
STATIC XML MAP



DYNAMIC ROUTING



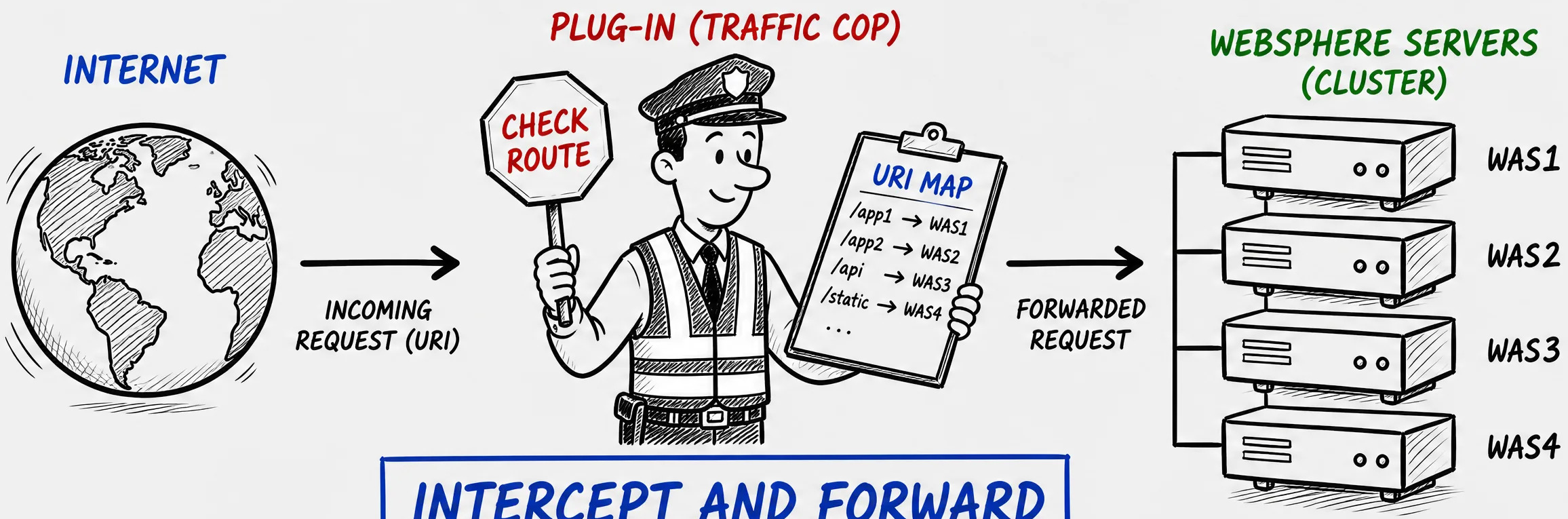
REAL-TIME AWARENESS
ADAPT. ROUTE. DELIVER.

IN REAL TIME:
SYNC & ADAPT

☆ Dynamic routing replaces static files with real-time awareness.

The Request Routing Flow

ProTech
protechtraining.com



INTERCEPT AND FORWARD

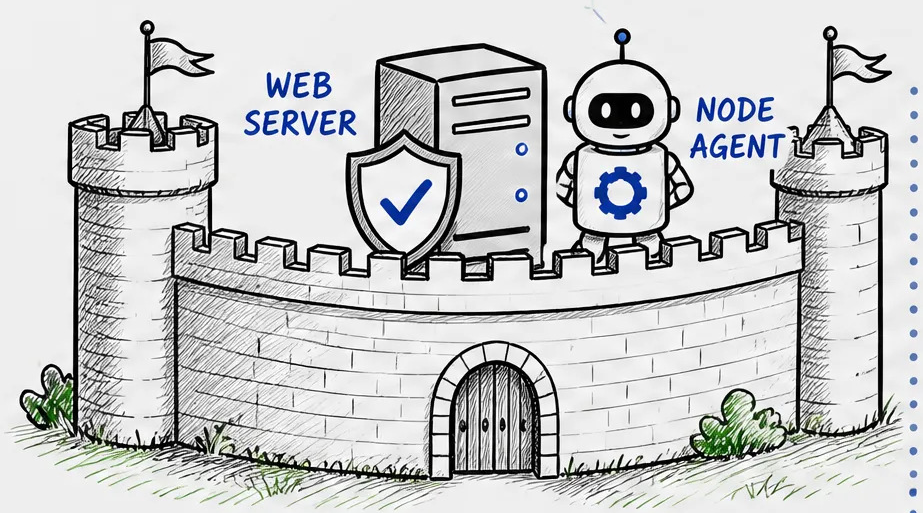
☆ The plug-in is a traffic cop reading a very specific map.

ProTech
protechtraining.com

Managed vs. Unmanaged Nodes

Dr. Lee

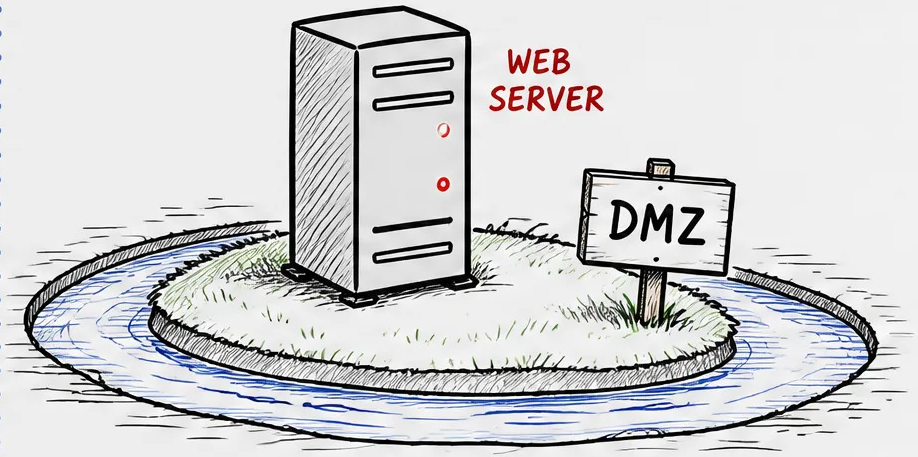
MANAGED NODE
(PROTECTED & MANAGED)



- ☑ Inside secure environment
- ☑ Monitored by Node Agent
- ☑ Policies enforced
- ☑ Fully managed and protected

LOCATION DICTATES MANAGEMENT

UNMANAGED NODE
(EXPOSED & UNMONITORED)

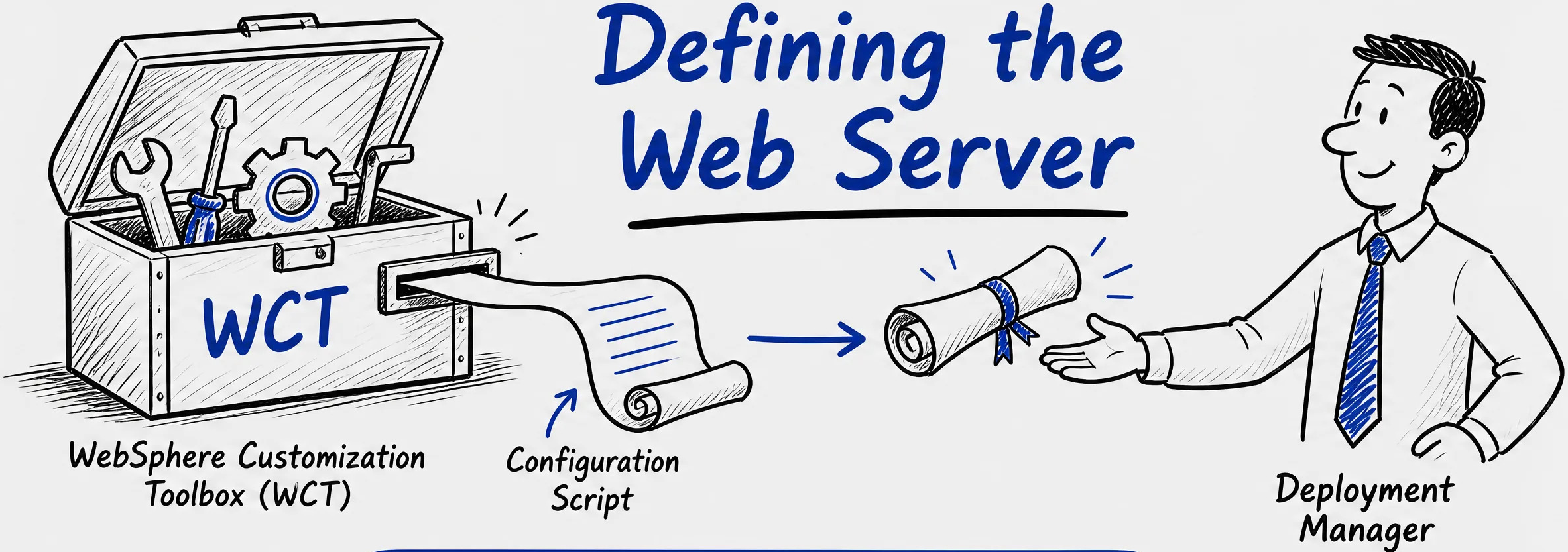


- ☒ Outside secure perimeter
- ☒ No Node Agent
- ☒ No visibility or enforcement
- ☒ High risk and unprotected

☆ **Location** dictates **management** strategy.

WebSphere Customization Toolbox

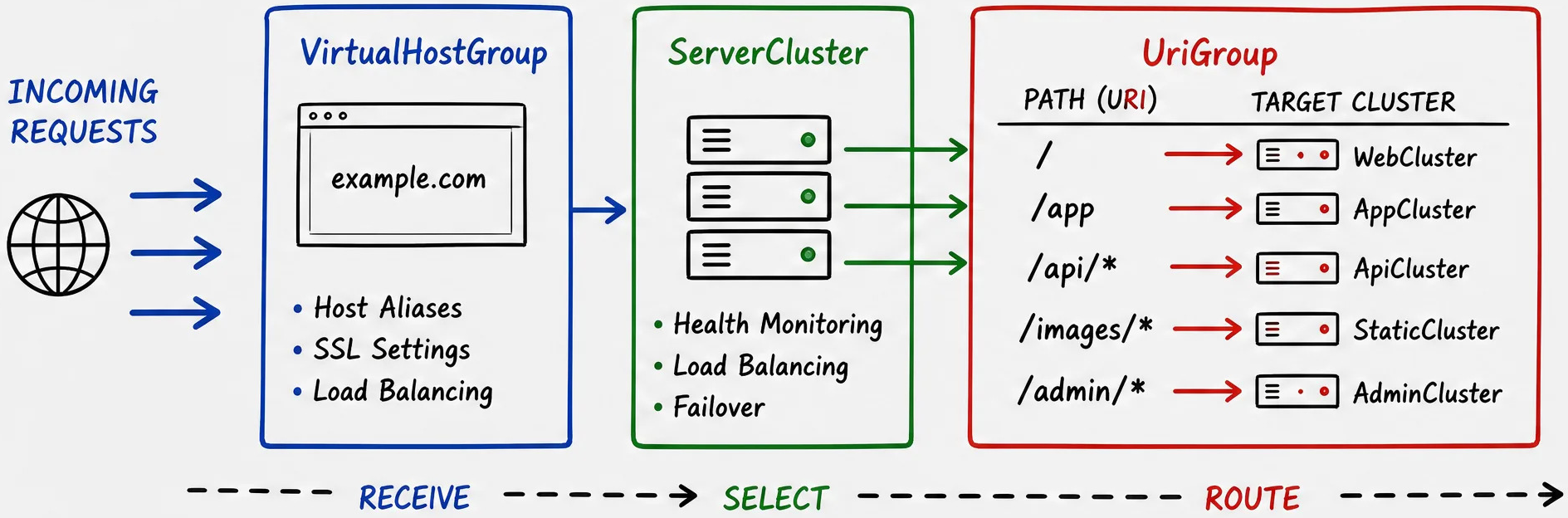
Defining the Web Server



☆ The toolbox writes the definition; the script executes the definition.

The plugin-cfg.xml File

The Static Topology Map



The XML file is the contract between the front door and the kitchen.

The plugin-cfg.xml file defines the complete routing and cluster topology used by the IBM HTTP Server plugin to handle requests.

Regeneration Triggers

Triggers

- App Install
- Virtual Host Edit
- Cluster Change

When the Map Changes



Stale plug-in files cause **404 errors** for running applications.

Propagation to the Web Server

Delivery of the Contract

GENERATION (Dmgr)

Dmgr writes (prints) the map (contract)



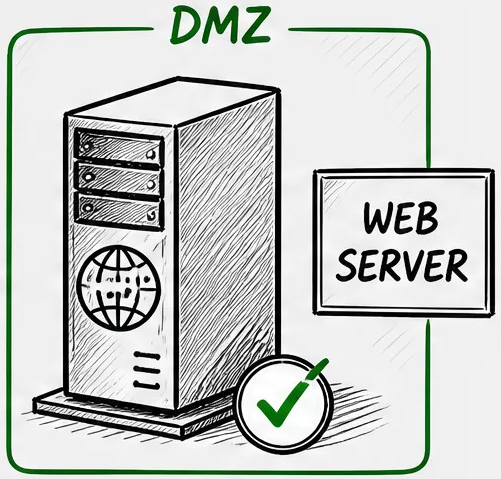
Dmgr prints the map (the contract).



Propagation delivers the map (the contract) across the network.

PROPAGATION (Destination)

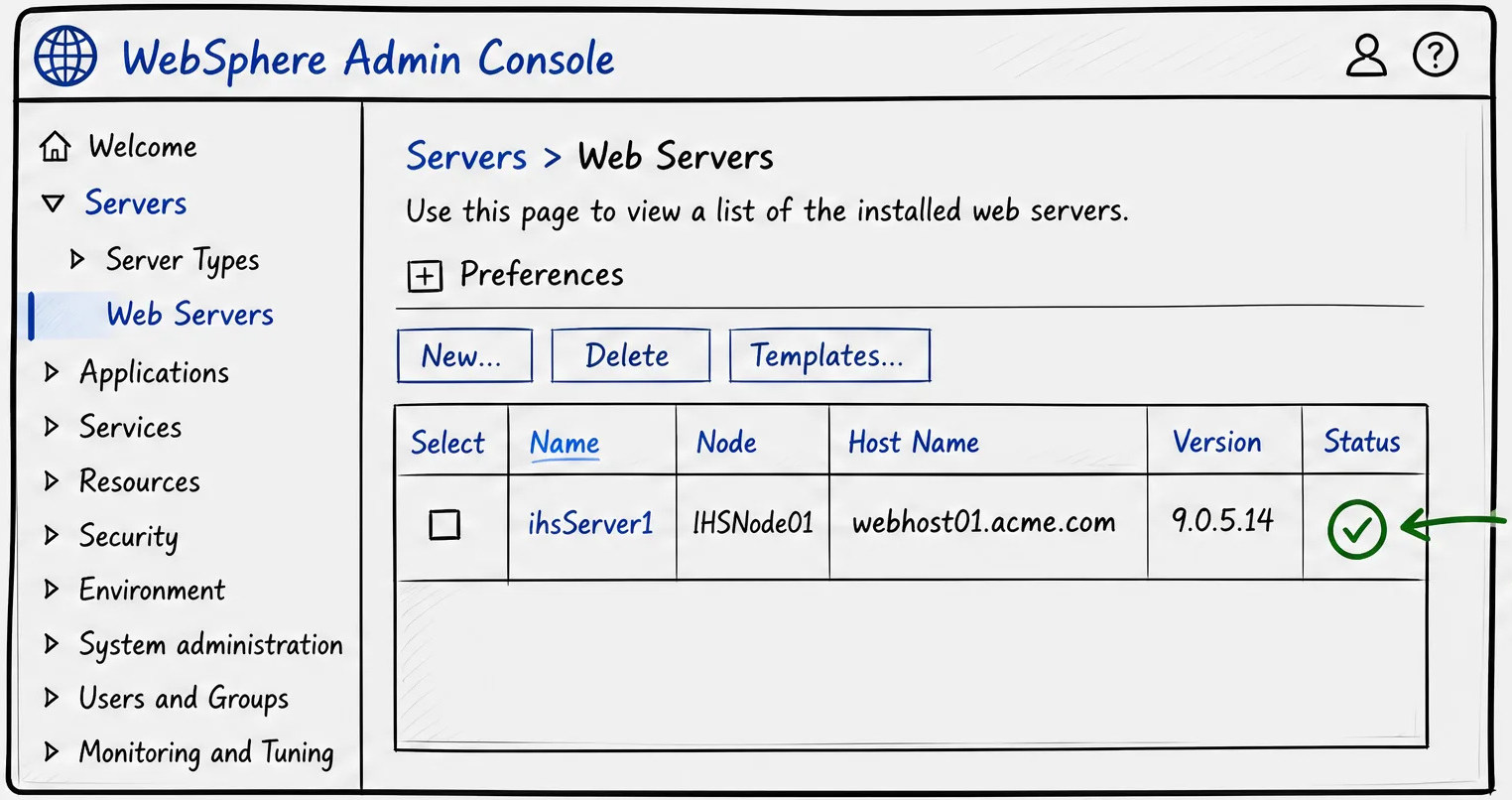
Map (contract) delivered to Web Server

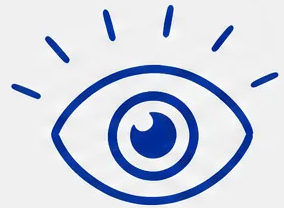


Web Server (in DMZ) receives the contract.

☆ **Generation** is writing the map; **propagation** is delivering it.

Defining the Web Server in the Console





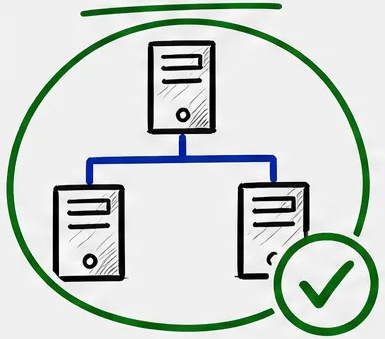
Centralized Visibility

View and manage all web servers in one central place for faster operations and smarter decisions.

Centralized visibility accelerates troubleshooting.

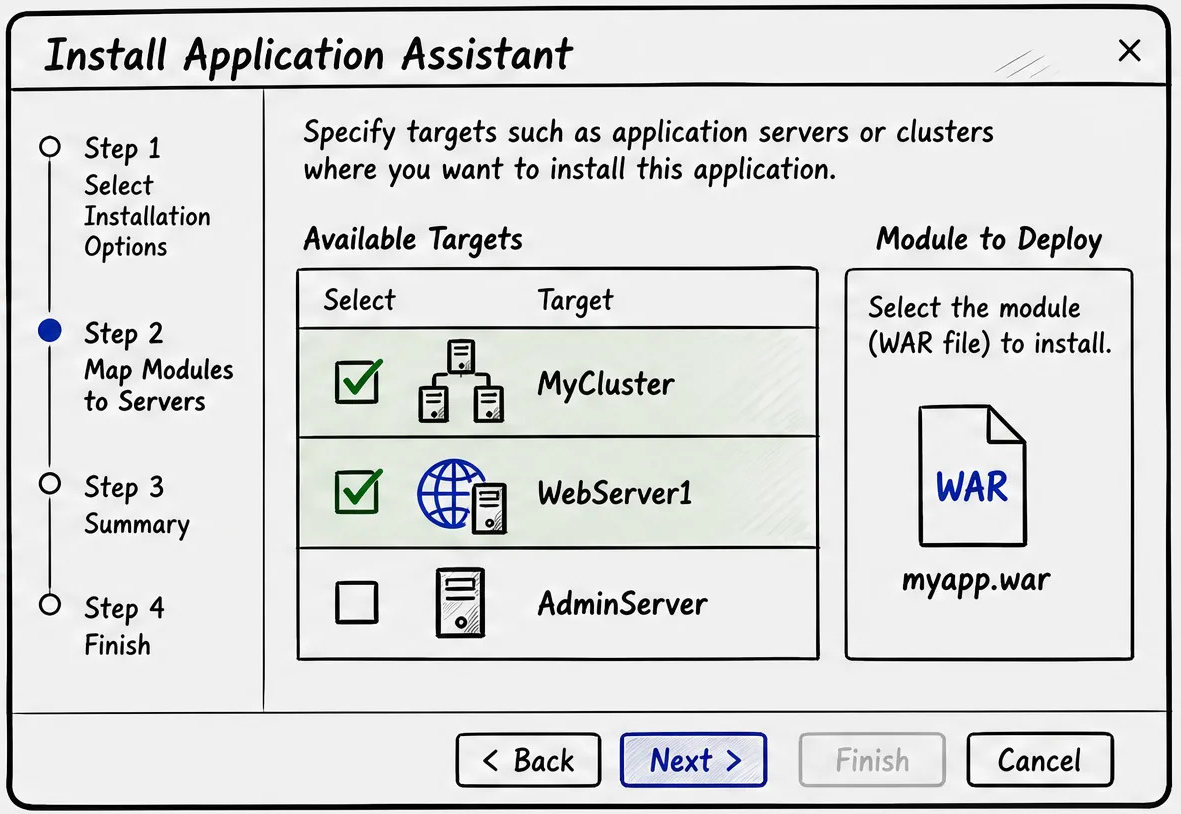
Mapping Modules to the Web Server

TARGET 1: CLUSTER



TARGET 2: WEB SERVER

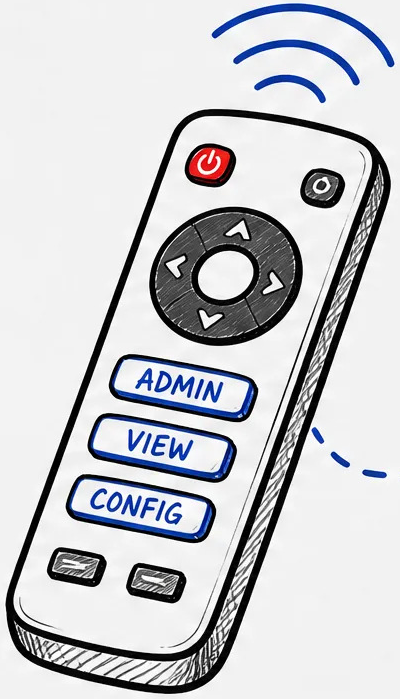




EXPLICIT PERMISSION TO ROUTE

If a module isn't mapped to the web server, the plug-in won't route traffic to it.

IHS Remote Administration



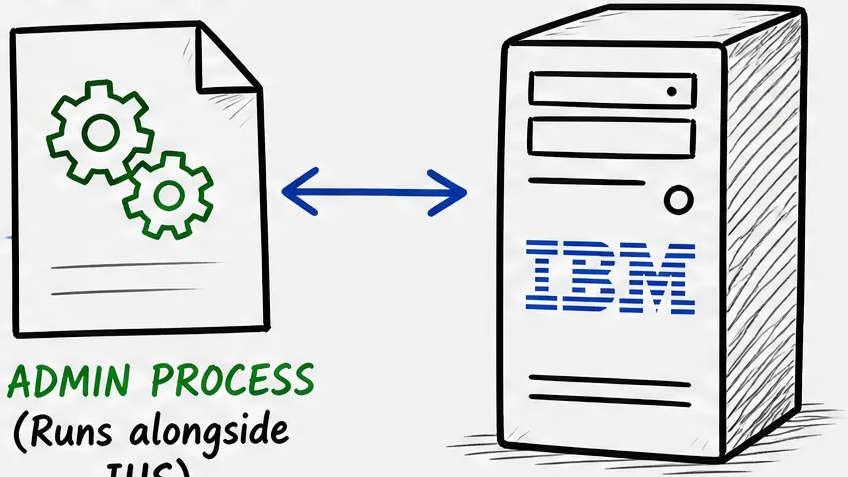
REMOTE ADMIN
(ANYWHERE)

CONTROLLING THE UNMANAGED

OUT IN THE DMZ

SPECIAL
admin.conf

IBM
HTTP SERVER



ADMIN PROCESS
(Runs alongside
IHS)

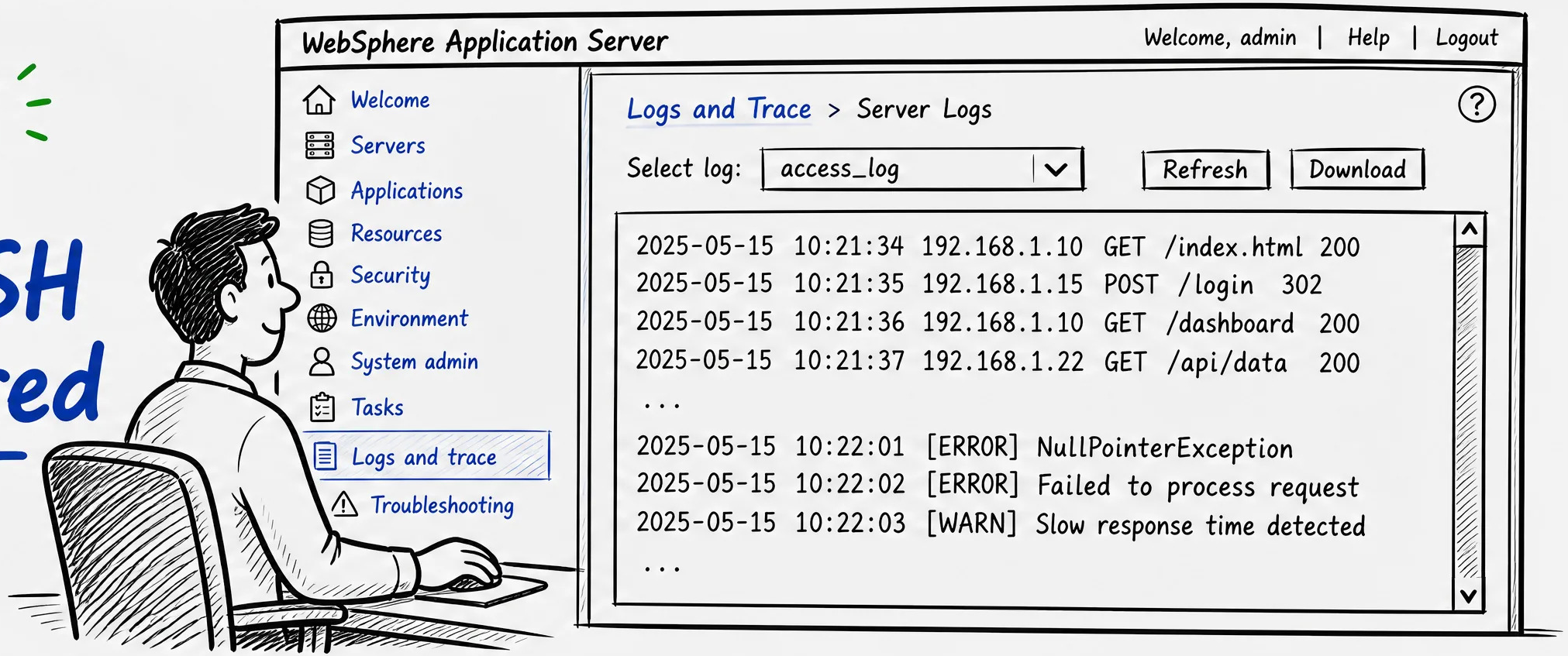
MEMORABLE STATEMENT:

Remote admin bridges the gap across unmanaged nodes.

Viewing Web Server Logs



No SSH Required



☆ Console log viewing prevents unnecessary SSH access.

Troubleshooting: Enable Plug-in Trace

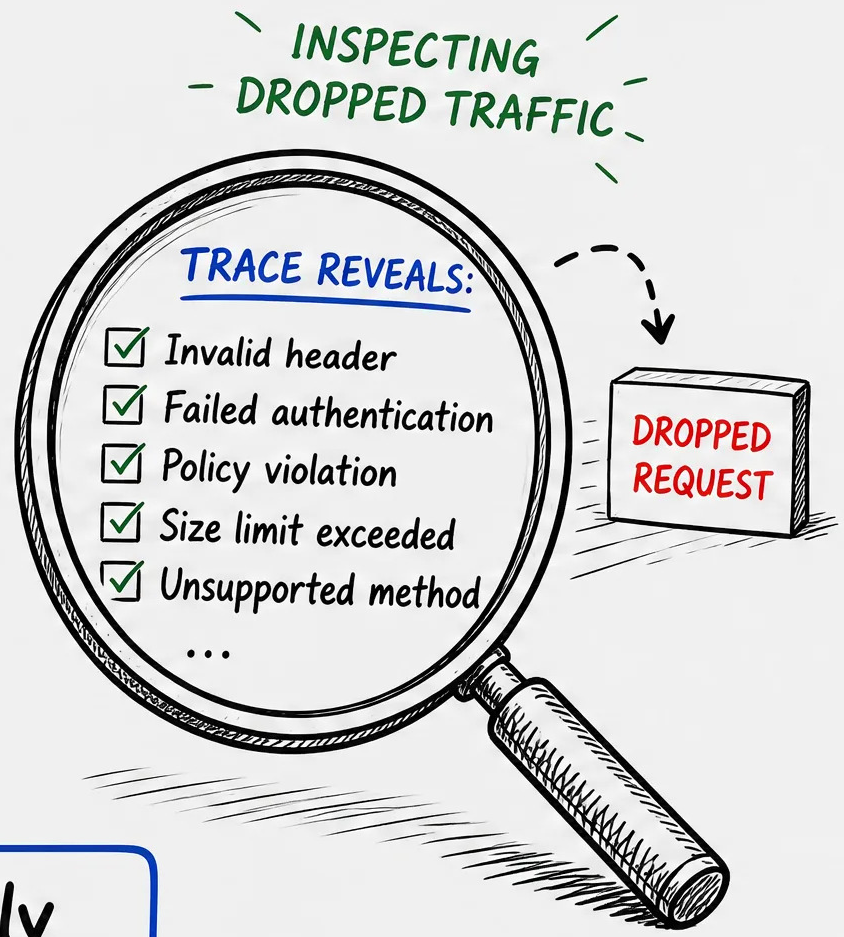
plugin-cfg.xml

```
<PluginConfiguration>
  <TraceOptions>
    <Log
      LogLevel="Error" />
  </TraceOptions>
</PluginConfiguration>
```

CHANGE TO

```
<PluginConfiguration>
  <TraceOptions>
    <Log
      LogLevel="Trace" />
  </TraceOptions>
</PluginConfiguration>
```

Why Was it Dropped?

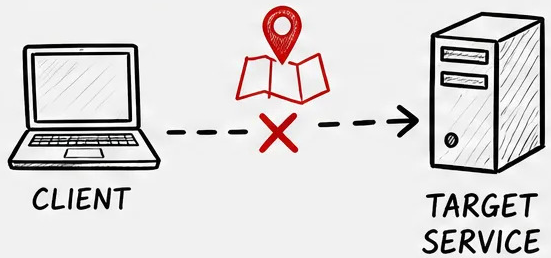


☆ The plug-in trace reveals exactly why a request was dropped.

Troubleshooting: Common Errors

ERROR #1

NO ROUTE FOUND (BAD MAPPING)

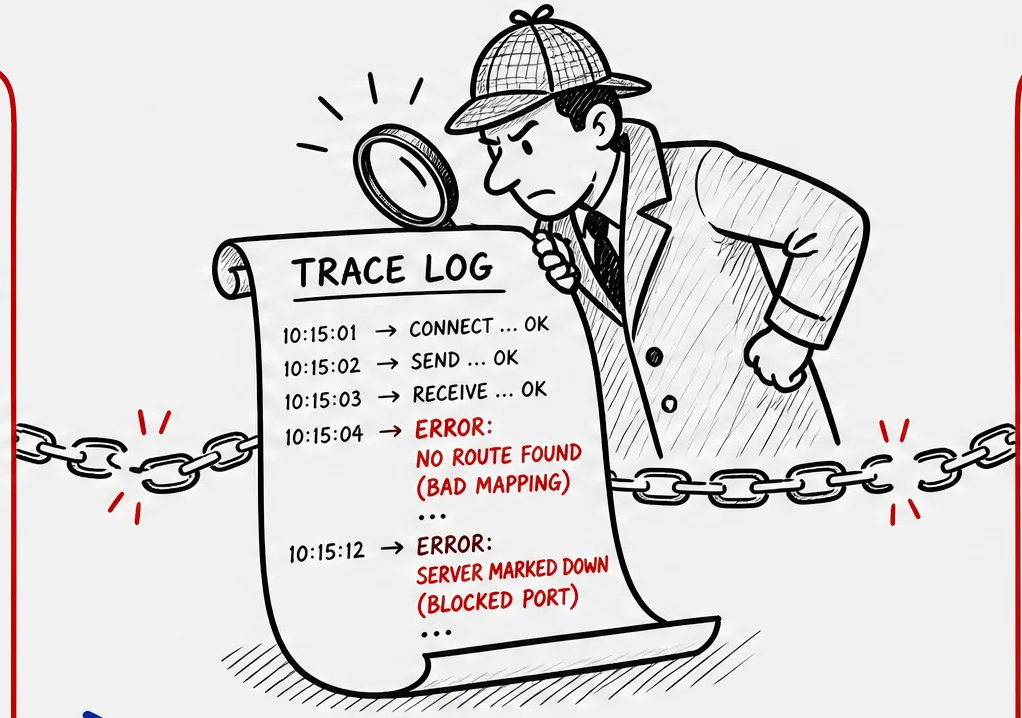


WHAT IT MEANS:

The request can't reach the intended destination.

CHECK:

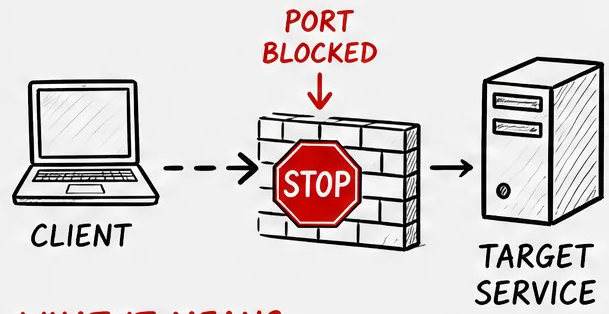
- Hostname / IP mapping
- DNS resolution
- Network routing / gateways



IDENTIFY THE BROKEN LINK

ERROR #2

SERVER MARKED DOWN (BLOCKED PORT)



WHAT IT MEANS:

The server is unreachable on the specified port.

CHECK:

- Port number / service status
- Firewall / security rules
- Server process availability

☆ Let the **trace log** identify the **broken link** in the chain.

The Impact of File Serving



Avoid the Wildcard Trap

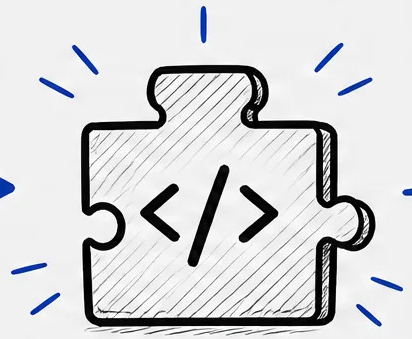
⚠ Wildcards destroy web server caching.

Module 5 Summary

1
MAP MODULE



2
GENERATE PLUG-IN



Create the plug-in (Route Definition)

3
PROPAGATE FILE





The Golden Rule of Routing

> “Map the module, Generate the plug-in, Propagate the file.”